

Packages

Are you struggling?

How people think programming looks



How it really looks 90% of the time



Plan for today

- Brief recap
- **Packages**

Recap of using variables

- Most of the time, we want to do more than add, subtract, multiply etc.
- We want to act on our variables. We do this with operators & **functions**
 - *Each function* has a unique name
 - *Each function* requires some input, and the function can be modified using arguments
 - *Each function* will produce an output
- Remember:
 - objects are *nouns*
 - functions & operators are *verbs*
 - arguments are *adverbs*

Recap of using variables

Where do you find functions?

- Some exist in R by default
 - `t.test()`
 - `cor()`
 - `scale()`

Lots of people all over the world write their own functions. And they (rightly!) think it's useful to share these functions.

Packages

What is a **package**?

- A collection of functions and datasets
- Open source

Packages are the reason R is so powerful!

- While you can definitely write your own functions, most of what you need to do someone else has already done for you!

Files Plots Packages Help Viewer				
 Install  Update  Packrat 				
Name	Description	Ver...		
System Library				
☐ abind	Combine Multidimensional Arrays	1.4-5		
☐ acepack	ACE and AVAS for Selecting Multiple Regression Transformations	1.4.1		
☐ arm	Data Analysis Using Regression and Multilevel/Hierarchical Models	1.11-1		
☐ AsioHea...	'Asio' C++ Header Files	1.12.2-1		
☐ askpass	Safe Password Entry for R, Git, and SSH	1.1		
☐ assertthat	Easy Pre and Post Assertions	0.2.1		
☐ backports	Reimplementations of Functions Introduced Since R-3.0.0	1.1.6		
☒ base	The R Base Package	4.0.0		
☐ base64...	Tools for base64 encoding	0.1-3		

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How do I get packages?

Packages can be downloaded from the *CRAN* (Comprehensive *R* Archive Network)

You will do this from inside [R](#)

Need to be connected to the internet!

2 ways to install packages

1. Install button in the Packages tab
2. R Code

Either way, you need to know the name of the package

Install Button

Install Packages

Install from: [? Configuring Repositories](#)

Repository (CRAN)

Packages (separate multiple with space or comma):

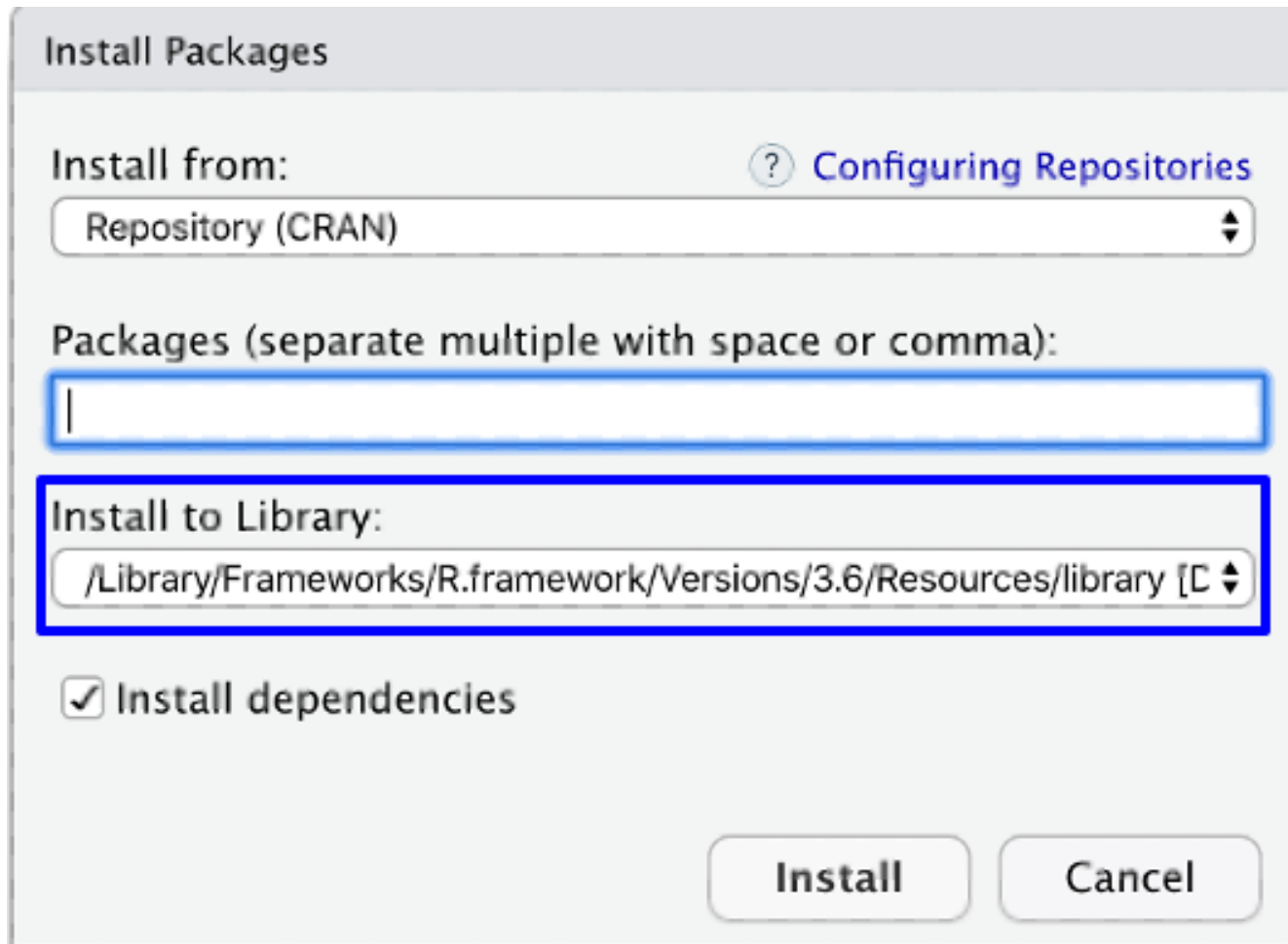
Install to Library:

/Library/Frameworks/R.framework/Versions/3.6/Resources/library [C]

☒ Install dependencies

Install Cancel

Install Button



The image shows the 'Install Packages' dialog box in RStudio. It has a title bar 'Install Packages'. Below the title bar, there is a section 'Install from:' with a help icon and a link 'Configuring Repositories'. Below this is a dropdown menu showing 'Repository (CRAN)'. Then, there is a section 'Packages (separate multiple with space or comma):' followed by an empty text input field. Below that is a section 'Install to Library:' followed by a text input field showing the default path '/Library/Frameworks/R.framework/Versions/3.6/Resources/library [C'. At the bottom, there is a checkbox 'Install dependencies' which is checked. At the very bottom, there are two buttons: 'Install' and 'Cancel'.

Install Packages

Install from: [? Configuring Repositories](#)

Repository (CRAN)

Packages (separate multiple with space or comma):

Install to Library:

/Library/Frameworks/R.framework/Versions/3.6/Resources/library [C

☒ Install dependencies

Install Cancel

Install Button

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☒ Install dependencies

Install Cancel

R code to install packages

```
install.packages("psych")
```

Packages

INSTALLING

- Downloading the package and saving it to your computer
- Like installing Microsoft Word on your computer
- Do this **ONCE**

LOADING

- Like opening Microsoft Word to write a paper
- Once a package is loaded, you can use all of it's functions and datasets are ready to use
- You need to do this **EVERY TIME** you open a new **R** session

2 ways to load packages

1. Checkbox in the packages tab (not recommended)
2. R code

Either way, you need to know the name of the package

R code to load packages

```
library(psych)
```

Dependencies

```
> library(lme4)  
Loading required package: Matrix  
Loading required package: Rcpp
```

Uses functions from other packages

Installed automatically

Loaded automatically

Help! (again)

Ways to find documentation:

`?psych` -- opens documentation specific to that package or function

`??psych` -- searches for this in all documentation (that you have installed and loaded)

To find a package that does what you need: **Google**

Summary

Packages are a collection of functions and data sets

1. You **install** the package once; must be connected to the internet
2. You **load** the package every time you use it; do not need to be connected to the internet

How do you find the function you need? How do you know what package it's in?

- **G-o-o-g-l-e!**
- "structural equation modeling in R"

How do you know how to use the function? What are the function's arguments?

- Help documentation in R
- **? function.name**